



CS 540 Introduction to Artificial Intelligence

Natural Language Processing (before LLMs)

University of Wisconsin-Madison
Spring 2025

Announcements

- **HW 2 online:**
 - Deadline **today 11:59PM**
- **HW 3:**
 - Deadline **Thursday Feb 13th at 11:59PM**
 - PCA
- Class roadmap:

NLP

Machine Learning: Introduction

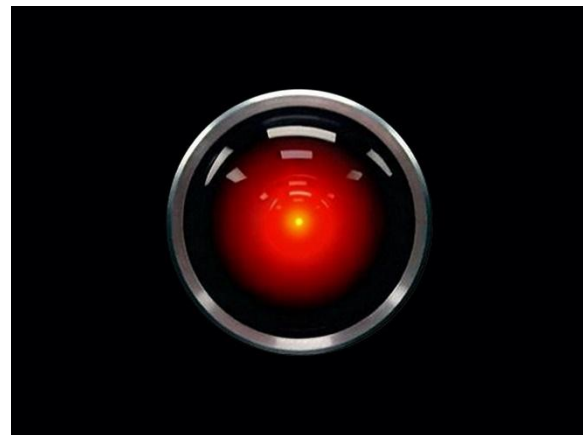
Machine Learning: Unsupervised
Learning I, II

Machine Learning: Linear regression

What is **NLP**?

Combining computing with human language. Want to:

- Answer questions
- Summarize or extract information
- Translate between languages
- Generate dialogue/language
- Write stories automatically



Why is it **hard**?

Many reasons:

- Ambiguity: “*Mary saw the duck with the telescope in the park*”. Several meanings.
- Understanding of the world
 - “Bob and Joe are fathers”.
 - “Bob and Joe are brothers”.



Approaches to NLP

A brief history

- Symbolic NLP: 50's to 90's
- Statistical/Probabilistic: 90's to present
 - Neural nets: 2010's to present
 - Large Language Model (LLM): GPT etc.

Lots of progress!

Lots more to work to do



ELIZA program

Outline

- Introduction to language models
 - n-grams, training, evaluation, generation
- Word representations
 - One-hot, word embeddings, transformer-based

Language Models

- ★ Basic idea: use probabilistic models to assign a probability to a sentence W

→ (predicting the next word given the existing data)

$$P(W) = P(w_1, w_2, \dots, w_n) \text{ or } P(w_{\text{next}} | w_1, w_2 \dots)$$

↓
sentence

↓
word

- Goes back to Shannon
 - Information theory: letters

Zero-order approximation	XFOML RXKHRJFFJUJ ALPWXFJWJXYJ FFJEYVJCQSGHYD QPAAMKBZAACIBZLKJQD
First-order approximation	OCRO HLO RGWR NMIELWS EU LL NBNESEBYA TH EEI ALHENHTTPA OObTTVA NAH BRL
Second-order approximation	ON IE ANTSOUTINY'S ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TUCOOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE
Third-order approximation	IN NO IST LAT WHEY CRATICT FROURE BIRS GROCID PONDENOME OF DEMONSTURES OF THE REPTAGIN IS REGOACTIONA OF CRE
First-order word approximation	REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NATURAL HERE HE THE A IN CAME THE TO OF TO EXPERT GRAY COME TO FURNISHES THE LINE MESSAGE HAD BE THESE

Training The Model

Recall the chain rule of probability:

$$P(w_1, w_2, \dots, w_n) = P(w_1)P(w_2|w_1) \dots P(w_n|w_{n-1} \dots w_1)$$


- How do we estimate these probabilities?
 - I.e., “training” in machine learning.
- From data (text corpus)
 - Can’t estimate reliably for long histories.

Training: Make Assumptions

- Markov assumption with shorter history:

$$P(w_i | w_{i-1} w_{i-2} \dots w_1) = P(w_i | w_{i-1} w_{i-2} \dots w_{i-k})$$

Markov chain



- Present doesn't depend on whole past

- Just recent past, i.e., *context*.
- What's ***k=0***?

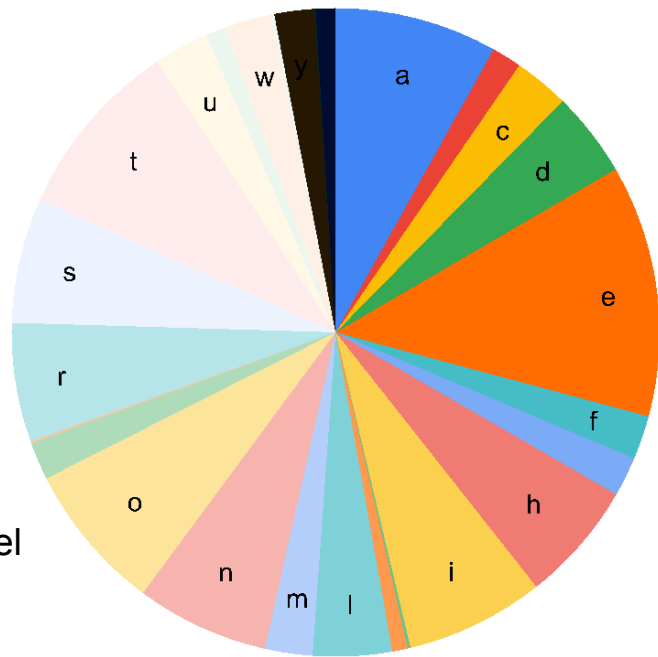
In this case, no history at all.

★ k=0: Unigram Model

- Full independence assumption:
 - (Present doesn't depend on the past)

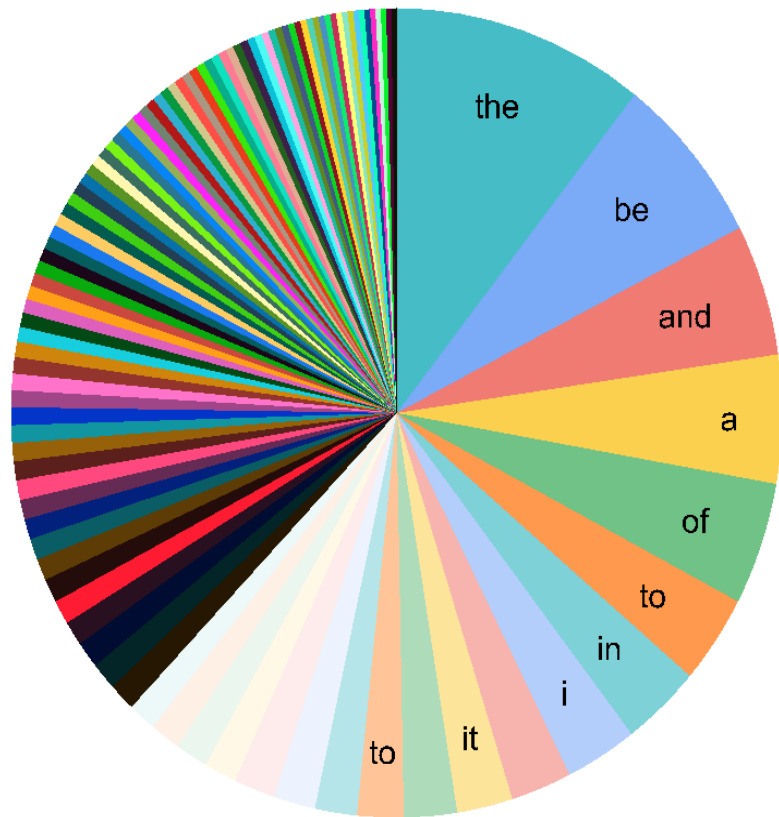
$$P(w_1, w_2, \dots, w_n) = P(w_1)P(w_2) \dots P(w_n)$$

(★ Future word is totally independent to the current word)



The English letter frequency wheel

Unigram word model



Example (from Dan Jurafsky's notes)

fifth, an, of, futures, the, an, incorporated, a,
a, the, inflation, most, dollars, quarter, in, is,
mass thrift, did, eighty, said, hard, 'm, july,
bullish that, or, limited, the

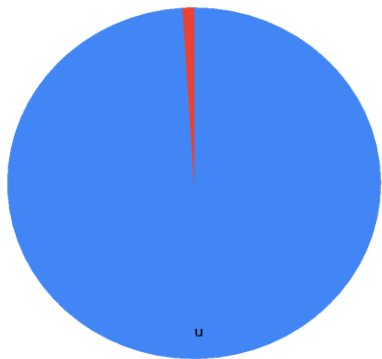
k=1: **Bigram Model**



• Markov Assumption:

- (Present depends on immediate past)

$$P(w_1, w_2, \dots, w_n) = P(w_1)P(w_2|w_1)P(w_3|w_2) \dots P(w_n|w_{n-1})$$



$p(.|q)$: the “after q” wheel



$p(.|j)$: the “after j” wheel

texaco, rose, one, in, this, issue,
is, pursuing, growth, in, a, boiler,
house, said, mr., gurria, mexico, 's,
motion, control, proposal, without,
permission, from, five, hundred,
fifty, five, yen outside, new, car,
parking, lot, of, the, agreement,
reached this, would, be, a, record,
november

k=n-1: n-gram Model

Can do trigrams, 4-grams, and so on

- More expressive as n goes up
- Harder to estimate

Training: just count? I.e, for bigram:

$$P(w_i | w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i)}{\text{count}(w_{i-1})}$$

↑ the count of w_{i-1} and w_i pairs

↓ The count of w_{i-1}

Simple “generative AI” from letter bigram (Markov Chain)

Writing = sampling

- Say we start with q
- Sample from $P(\cdot \mid q)$: spin the “after q” wheel  , we get u
- Sample from $P(\cdot \mid u)$: spin the “after u” wheel, say we get e
- Sample from $P(\cdot \mid e)$: spin the “after e” wheel, say we get r
- ...

Sampling Shakespeare unigram LM

- To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have
- Every enter now severally so, let
- Hill he late speaks; or! a more to leg less first you enter
- Will rash been and by I the me loves gentle me not slavish page, the and hour; ill let
- Are where exeunt and sighs have rise excellency took of .. sleep knave we. near; vile like

Sampling Shakespeare bigram LM

- What means, sir. I confess she? then all sorts, he is trim, captain.
- Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow.
- What we, hath got so she that I rest and sent to scold and nature bankrupt, nor the first gentleman?
- Enter Menenius, if it so many good direction found'st thou art a strong upon command of fear not a liberal largess given away, Falstaff! Exeunt

Sampling Shakespeare trigram LM

- Sweet prince, Falstaff shall die. Harry of Monmouth's grave.
- This shall forbid it should be branded, if renown made it empty.
- What ist that cried?
- Indeed the duke; and had a very good friend.

n-gram Training

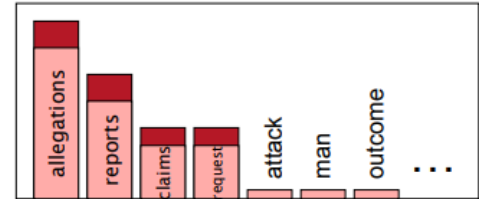
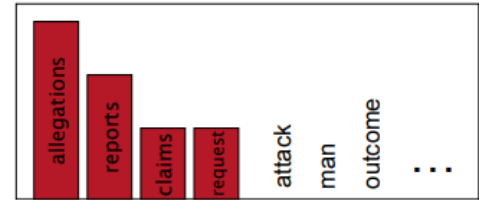
Basically just determine these conditional probabilities

Issues:

$$P(w_i|w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i)}{\text{count}(w_{i-1})}$$

- 1. Multiply tiny numbers?
 - **Solution:** use logs; add instead of multiply
- 2. n-grams with zero probability?
 - **Solution:** smoothing

$P(w|\text{denied the})$



$$P(w_i|w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i) + 1}{\text{count}(w_{i-1}) + V}$$

$\sum_{w_i \in V} P(w_i|w_{i-1}) = 1$

★ V times
the length of the vocabulary

Break & Quiz

Q 1.1: Which of the below are bigrams from the sentence “It is cold outside today”.

- A. It is
- B. cold today
- C. is cold
- D. A & C

Break & Quiz

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- **D. A & C**

Break & Quiz

Q 1.2: Smoothing is increasingly useful for n-grams when

- A. n gets larger
- B. n gets smaller
- C. always the same
- D. n larger than 10

Break & Quiz

Q 1.2: Smoothing is increasingly useful for n-grams when

- **A. n gets larger**
- B. n gets smaller
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Evaluating Language Models

How do we know we've done a good job?

- Observation
- Train/test on separate data & measure metrics



Metrics:

- 1. Extrinsic evaluation
- 2. Perplexity



★ Extrinsic Evaluation

How do we know we've done a good job?

- Pick a task and use the model to do the task
- For two models, M_1 , M_2 , compare the accuracy for each task
 - Ex: Q/A system: how many questions right. Translation: how many words translated correctly
- Downside: slow; may change relatively

Detect language



English

Enter text

Translation

★ Intrinsic Evaluation: Perplexity

Perplexity is a **measure of uncertainty**

$$PP(W) = P(w_1, w_2, \dots, w_n)^{-\frac{1}{n}}$$

Compute average $PP(W)$ for all W from a dataset

★ **Lower is better!** Examples:

- WSJ corpus; 40 million words for training:
 - Unigram: 962, Bigram 170, Trigram 109

Further NLP Tasks

Language modeling is **not the only NLP task**:

- Part-of-speech tagging, parsing, etc.
- Question-answering, translation, summarization, classification (e.g., sentiment analysis), generation, etc.

Break & Quiz

Q 2.1: What is the perplexity for a sequence of n digits 0-9? All occur independently with equal probability.

- A. 10
- B. 1/10
- C. 10^n
- D. 0

$$\text{PP}(W) = P(w_1, w_2, \dots, w_n)^{-\frac{1}{n}}$$

unigrams (k=1)

No past needed: means $P(w_1 \dots w_n) = P(w_1) P(w_2) \dots P(w_n)$

$$= \left[\frac{1}{10} \right]^n \cdot \frac{1}{n} = 10$$

Break & Quiz

Q 2.1: What is the perplexity for a sequence of n digits 0-9? All occur independently with equal probability.

- **A. 10**
- B. $1/10$
- C. 10^n
- D. 0

$$\text{PP}(W) = P(w_1, w_2, \dots, w_n)^{-\frac{1}{n}}$$

$$(P(w_1) * P(w_2) \dots * P(w_{10}))^{(-1/10)} = ((1/10) * (1/10) * \dots * (1/10))^{(-1/10)} = 10$$

Representing Words

Remember value of random variables (**RVs**)

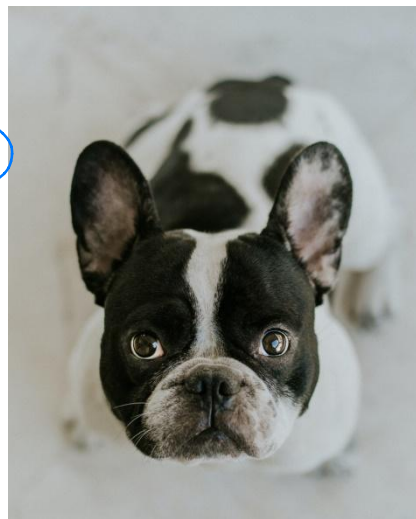
- Easier to work with than objects like 'dog'

Traditional representation: **one-hot vectors**

$$\text{dog} = [0 \ 0 \ 0 \ 0 \ 1 \ 0]$$

- Dimension: # of words in vocabulary
- Relationships between words?

★ (Each word is a basis vector with one 1 in it)



Smarter Representations

Distributional semantics: account for relationships

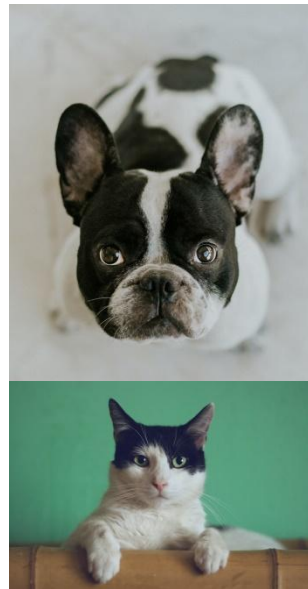
- Reps should be close/similar to other words that appear in a similar context

Dense vectors:

$$\text{dog} = [0.13 \quad 0.87 \quad -0.23 \quad 0.46 \quad 0.87 \quad -0.31]^T$$

$$\text{cat} = [0.07 \quad 1.03 \quad -0.43 \quad -0.21 \quad 1.11 \quad -0.34]^T$$

AKA **word embeddings**



Training Word Embeddings

Many approaches (super popular 2010-present)

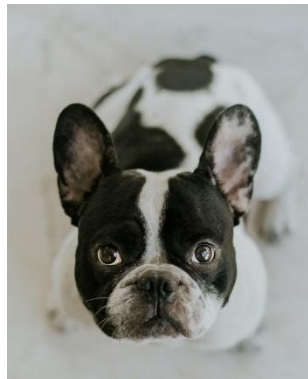
- Word2vec: a famous approach
- What's our likelihood?

Windows of length
2a

$$L(\theta) = \prod_{t=1}^T \prod_{-a \leq j \leq a} P(w_{t+j} | w_t, \theta)$$

Our word vectors

All positions



Training Word Embeddings

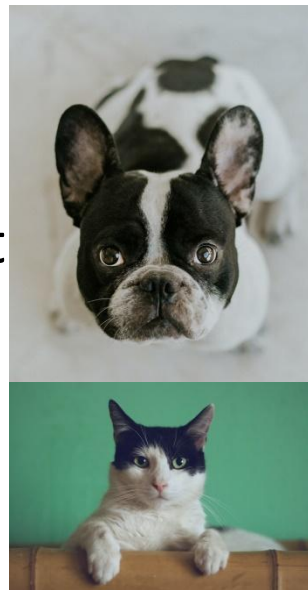
Word2vec likelihood

$$L(\theta) = \prod_{t=1}^T \prod_{-a \leq j \leq a} P(w_{t+j} | w_t, \theta)$$

- Maximize this; what's the probability?
 - Two vectors per word. v_w , u_w for center/context (o is context word, c is center)

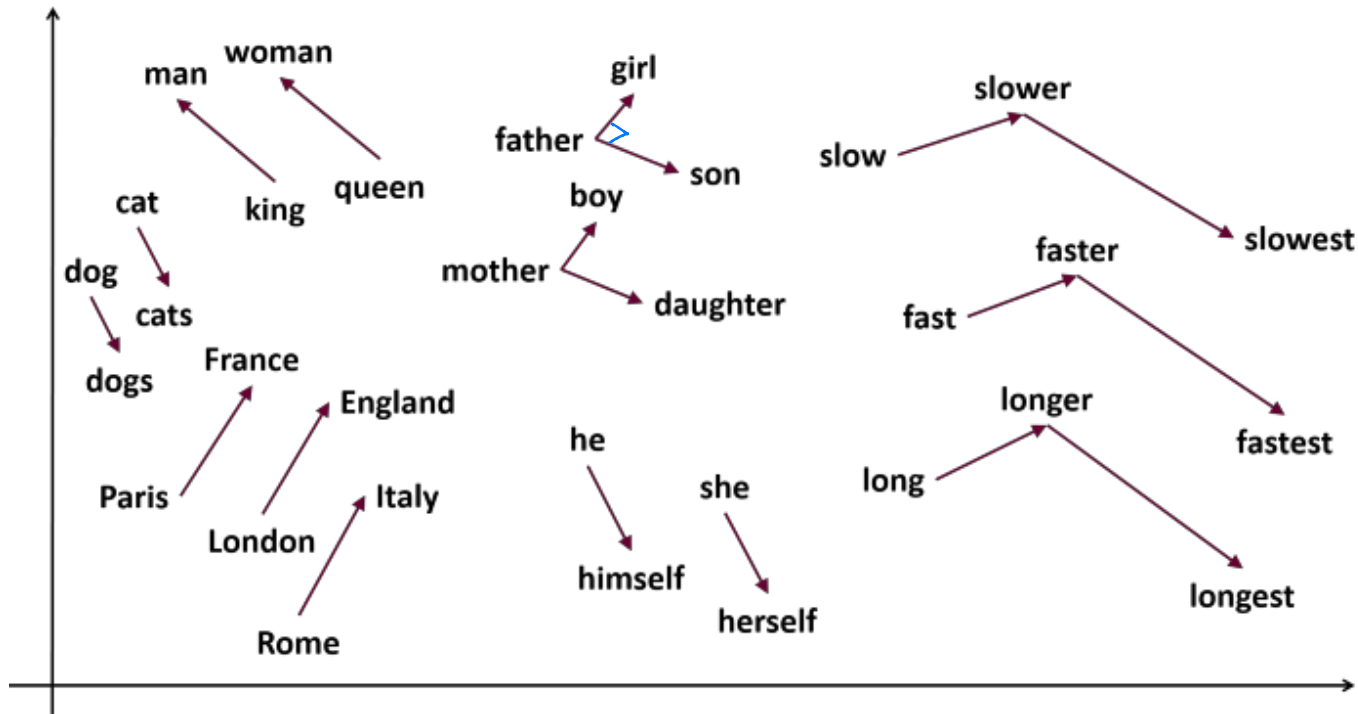
Similarity $\longrightarrow$

$$P(o|c) = \frac{\exp(u_o^T v_c)}{\sum_{w \in V} \exp(u_w^T v_c)}$$



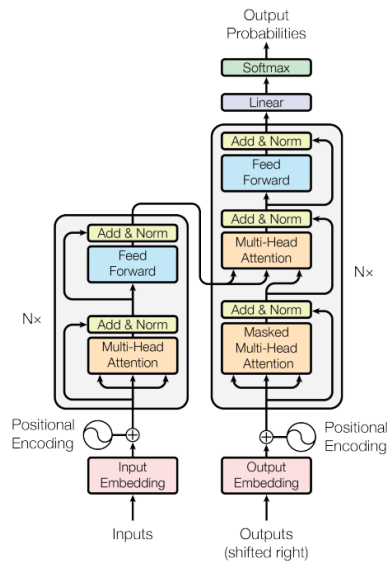
Word Embeddings

★ (The larger the angle is, the more different two words are)



Beyond “Shallow” Embeddings

- Transformers: special model architectures based on attention
 - Sophisticated types of neural networks
- Pretrained models
 - Based on transformers: BERT, GPT
 - Include context!
- **Fine-tune** for desired task



Reading

- Natural Language and Statistics, Notes by Zhu.
<https://pages.cs.wisc.edu/~jerryzhu/cs540/handouts/NLP.pdf>